Business Understanding Report

Project: Heart Disease Prediction | AnalystLab Africa – Data Science Internship Programme – Week 2

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1. The Business Problem

Cardiovascular diseases are among the leading causes of death worldwide. Hospitals and healthcare professionals need to quickly identify patients at high risk of developing heart disease so that appropriate preventive measures can be put in place. However, assessing this risk manually can be complex because of the large number of medical factors involved.

The organization therefore wants to leverage patients' medical data to improve the early identification of individuals at risk.

2. Project Objectives

The main objective of the project is to develop a dataset ready for Machine Learning in order to predict the presence or absence of heart disease in a patient.

The specific objectives are:

- Clean and prepare the medical data.
- Transform the variables so that they can be used by Machine Learning models.
- Identify the factors that most strongly influence the risk of heart disease.
- Build a reliable foundation for training a predictive model in the future.
- Help healthcare professionals make faster and more accurate decisions.

3. The Target Variable

The target variable is:

HeartDisease

This variable indicates whether or not the patient suffers from heart disease:

- 1 = Presence of heart disease.
- 0 = Absence of heart disease.

This is the variable the future model will need to predict from the other patient characteristics, such as age, sex, blood pressure, cholesterol, or medical examination results.

4. The Importance of Predictive Modelling

Predictive modelling uses historical data to estimate the probability that an event will occur in the future.

In the medical field, it offers several advantages:

- Early detection of patients at risk.
- Improved quality of care.
- Reduction of human assessment errors.
- Support for clinical decision-making.
- Optimized use of medical resources.

With Machine Learning, it becomes possible to analyze several risk factors simultaneously and uncover relationships that are difficult to identify manually.

5. Expected Benefits for the Organization

Implementing a heart disease prediction system can bring several benefits:

For hospitals and health centres

- Rapid identification of high-risk patients.
- Prioritization of medical examinations.
- Reduction of costs linked to cardiac complications.
- Better care planning.

For healthcare professionals

- Data-driven decision support.
- Better risk assessment.
- Time savings during preliminary diagnosis.

For patients

- Earlier diagnosis.
- Better prevention of cardiovascular diseases.
- Reduced risk of serious complications.
- Improved quality of life.

Conclusion

This project aims to prepare a medical dataset so that a model capable of predicting the presence of heart disease can be developed later. Such a solution can help improve medical decision-making, strengthen prevention, and optimize the care of patients at risk.